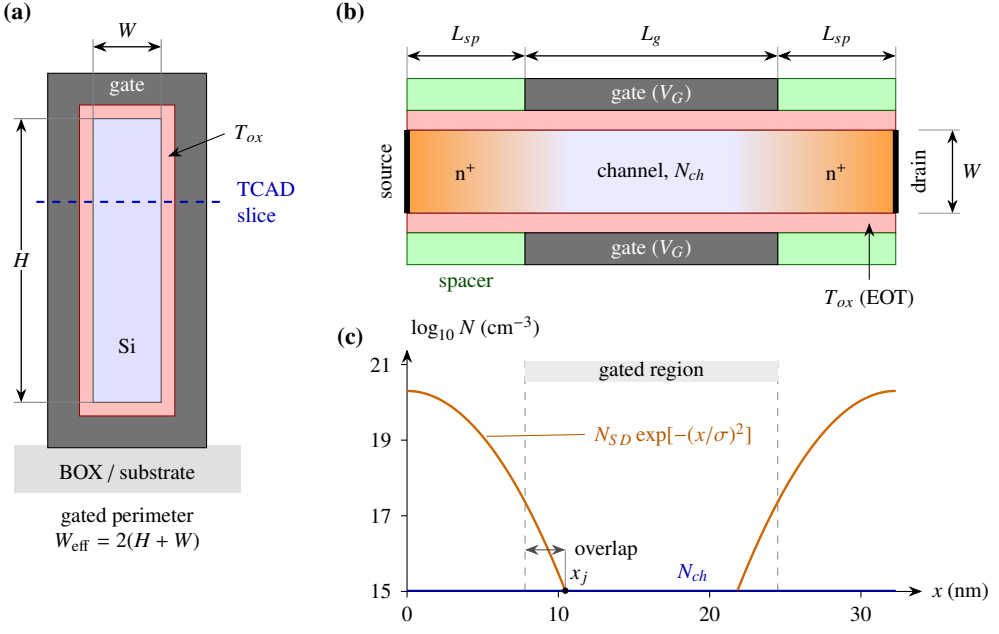
(a)
W
gate
T_{ox}
TCAD
slice
H
Si
BOX / substrate
gated perimeter
$W_{\text{eff}} = 2(H + W)$
(b)
L_{sp}
L_g
L_{sp}
gate (V_G)
source
n^+
channel, N_{ch}
n^+
drain
W
gate (V_G)
spacer
T_{ox} (EOT)
(c)
$\log_{10} N$ (cm^{-3})
21
19
17
15
gated region
$N_{SD} \exp[-(x/\sigma)^2]$
overlap
x_j
N_{ch}
0
10
20
30
x (nm)